Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Boolean conditions: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Translate stated requirements into **comparison**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate numeric expressions using Python’s grouping rules

Reference example, not the only acceptable wording: 3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Use fictional activities and **Boolean** operands throughout. Do not equate suitability with permission or claim that a programmed condition learns from data.

Reference for teacher precision: https://docs.python.org/3/library/stdtypes.html#**boolean**-operations-and-or-not. Python and/or may return operands of other types; this lesson deliberately confines them to **Boolean**s.

Outcomes

Translate stated requirements into **comparison**s

Combine **Boolean** values with and or and not

Choose a case that distinguishes two conditions

Preparation

Python 3 editor and conditions.py

True and False cards and written activity requirements

Blank **input**, separate checks, expected and observed columns

Prerequisite answer

False is a **Boolean** value; "False" is text. This week uses actual **Boolean** values for yes-or-no flags.

End goal for the pictorial case

Trace **boolean** conditions and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Boolean conditions using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Boolean conditions · 1

A fictional activity must take at most **12 minutes** and be quiet. An activity taking exactly **12 minutes** meets the time requirement. A quiet 15-minute activity does not.

Before writing code, underline each requirement and decide whether all requirements must hold or whether alternatives are allowed.

Explain Boolean conditions · 2

Use a fresh conditions.py file with nonnegative integer minutes and an actual **Boolean** quiet flag. A **comparison** produces True or False for these values. The expression minutes <= 12 includes 12; minutes < 12 excludes it.

Use == for equality and != for inequality. A single = assigns a value to a name; it does not ask whether two values are equal.

The script prints True then False. The first result concerns time only. The second requires time_ok and quiet. These are separate claims.

Trace each **input** before explaining the combined result; a result without its conditions can be misleading.

Term	Meaning
Comparison	An expression that checks a relation between values
Boolean	A type with the values True and False
Conjunction	A requirement that both conditions hold
Inclusive or	A condition satisfied when either or both parts hold
Counterexample	A case that shows a claim or rule is wrong

Two requirements must both hold

INPUT CASE	TIME CHECK	COMBINED RESULT
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

Inclusive or accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals <= changed to <. (8, False) should fail and reveals and changed to or.

Other **input**s are acceptable if they distinguish the versions.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A	B	A and B	A or B
False	False	False	False
False	True	False	True
True	False	False	True
True	True	True	True

Model the reasoning aloud · 3

With **Boolean** operands, and needs both parts True. **Inclusive or** needs at least one True, so both True is allowed. not reverses a **Boolean**: not True is False and not False is True.

Python can also apply and and or to other types; keep operands **Boolean** here so this table directly describes the result.

> **A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Boolean conditions

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

The facts we will use

> **Supplied facts**
>
> Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

AI ACADEMY / PYTHON-BRIDGE/4

Boolean conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check lower boundary

15>=12 is True.



2. Check upper boundary

15<=15 is also True.



3. Combine with and

Both comparisons are True, so allowed is True.

CHANGE ONE THING

Change age to 16.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check lower boundary	15>=12 is True.
2. Check upper boundary	15<=15 is also True.
3. Combine with and	Both comparisons are True, so allowed is True.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

What this example does not establish

Keep the stated limit

This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Change one thing in the pictured case

Change one thing

Change: Change age to 16.

Predicted result

Prediction: The result becomes False.

Why the result follows

Reason: The lower condition remains True, but the upper condition fails.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Boolean conditions, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Time or stage	Teacher action
Session 1 start to 7 minutes	Check Boolean versus text with the prerequisite.
7 to 20 minutes	Underline the two requirements and complete W1.
20 to 33 minutes	Model the diagram and trace the two printed values.
33 to 45 minutes	Complete W2 with predictions before running.
Session 2 start to 8 minutes	Retrieve an inclusive boundary and a one-condition failure.
8 to 20 minutes	Build the four-row truth table and complete W3 and the guided question.
20 to 35 minutes	Collect W4 independently, including a counterexample.
35 to 45 minutes	Discuss W5 and use the exit to diagnose boundary and logic errors.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use separate cards for each **comparison**, then combine them. Read at most and at least aloud with number lines.

Extension: explain why not ((minutes >= 5) and (minutes <= 9)) identifies out-of-range numeric values under the stated **input** contract.

Reduce help deliberately

Completion practice

Completion task: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the

time **comparison** and final suitability values before running the program.

Reasoning to listen for

Teacher check: Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: At most 12 includes 12; compare < and <= at exactly 12.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Reasoned teacher answer

Age 11 gives False/True, so False. Age 15 gives True/True, so True. Age 16 gives True/False, so False. Both interval boundaries must hold.

Changed-case reasoning: The result becomes False. The lower condition remains True, but the upper condition fails.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Boolean conditions investigation and use the first unsupported step to choose a teaching response.

At most excludes the boundary: At most 12 includes 12; compare < and <= at exactly 12.

or means exactly one: **Inclusive or** also accepts two True values.

A single equals sign tests equality: = assigns; == compares.

A True result proves a fact about the world: It establishes only that the supplied values satisfy this expression.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Boolean conditions questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Inclusive or accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals <= changed to <. (8, False) should fail and reveals and changed to or.

Other **input**s are acceptable if they distinguish the versions.

W1 Translate the requirement

Workbook question · W1

Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

Teacher reasoning · W1

(minutes <= 12) and quiet. At most includes equality, so <= is required; the nonnegative integer and **Boolean input** contract still applies.

W2 Trace separate checks

Workbook question · W2

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Teacher reasoning · W2

Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

W3 Distinguish alternatives

Workbook question · W3

Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

Teacher reasoning · W3

or gives False, True, True, True for FF, FT, TF, TT. Both available is accepted. not pencil is True when pencil is False and False when pencil is True.

W4 Write a new rule independently

Workbook question · W4

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

Choose a case that detects replacing and with or between the range check and available.

Teacher reasoning · W4

(minutes >= 5) and (minutes <= 9) and available. Results True, True, False, False. (7, False) distinguishes (range_check and available) from (range_check or available): correct False versus faulty True.

W5 Separate a check from a guarantee

Workbook question · W5

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

Teacher reasoning · W5

The expression checks supplied values against stated conditions. Inaccurate **input** can still pass. Permission is a separate **human decision** and must not be inferred from suitability.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires both inclusive bounds, the available flag, all four outcomes and a distinguishing case with reasoning.

After support, retry a 2-to-4-minute inclusive rule with an open flag: (2, True) passes, (4, False) fails and (5, True) fails. Record initial independent work before the retry.

Assess independent transfer

Independent question · withhold the key

Selected case: Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged.

Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

Selected case answer

Teacher key · Selected case answer

Original results: True, False, True, True. Full eligibility is False because availability is False. With history: False, False, True, True.

The final OR remains True because90<=100; it does not establish that the science requirement passed. NOT available is True in both traces.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect the project, support and next lesson

> Connect Boolean conditions to Python readiness portfolio, adapt support and plan a later changed-case check.

Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use separate cards for each **comparison**, then combine them. Read at most and at least aloud with number lines.

Extension: explain why not ((minutes >= 5) and (minutes <= 9)) identifies out-of-range numeric values under the stated **input** contract.

Week 5 uses conditions in if elif and else branches, then tests exact boundaries.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/5, python-bridge/7, python-bridge/10. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

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Prepare, teach and assess

> Use this preparation summary alongside the detailed Boolean conditions lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Translate stated requirements into **comparison**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Supported practice answer

Reasoned practice answer: Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

Check every learner before discussion: Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: At most 12 includes 12; compare < and <= at exactly 12.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change age to 16. Require a **prediction** and a reason before comparing. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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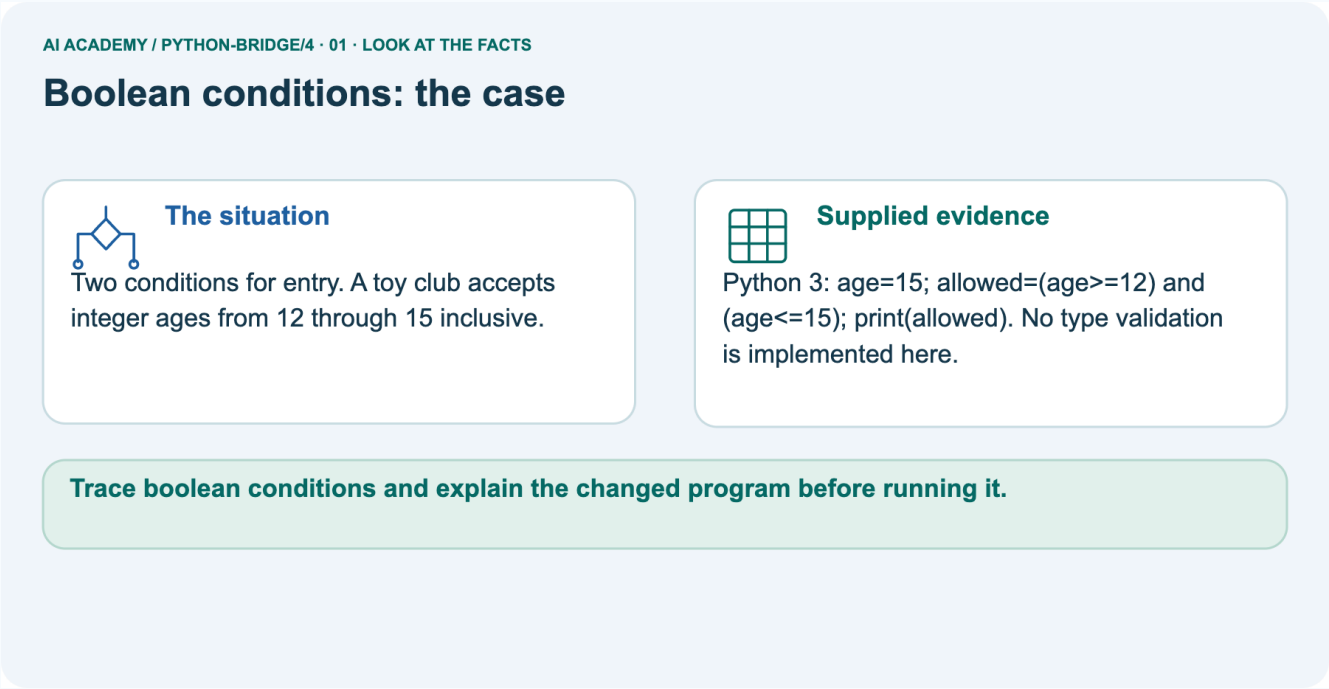
Visual concept journey

> Show one Boolean conditions graphic at a time, request a prediction and discuss its trace

> before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Boolean conditions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

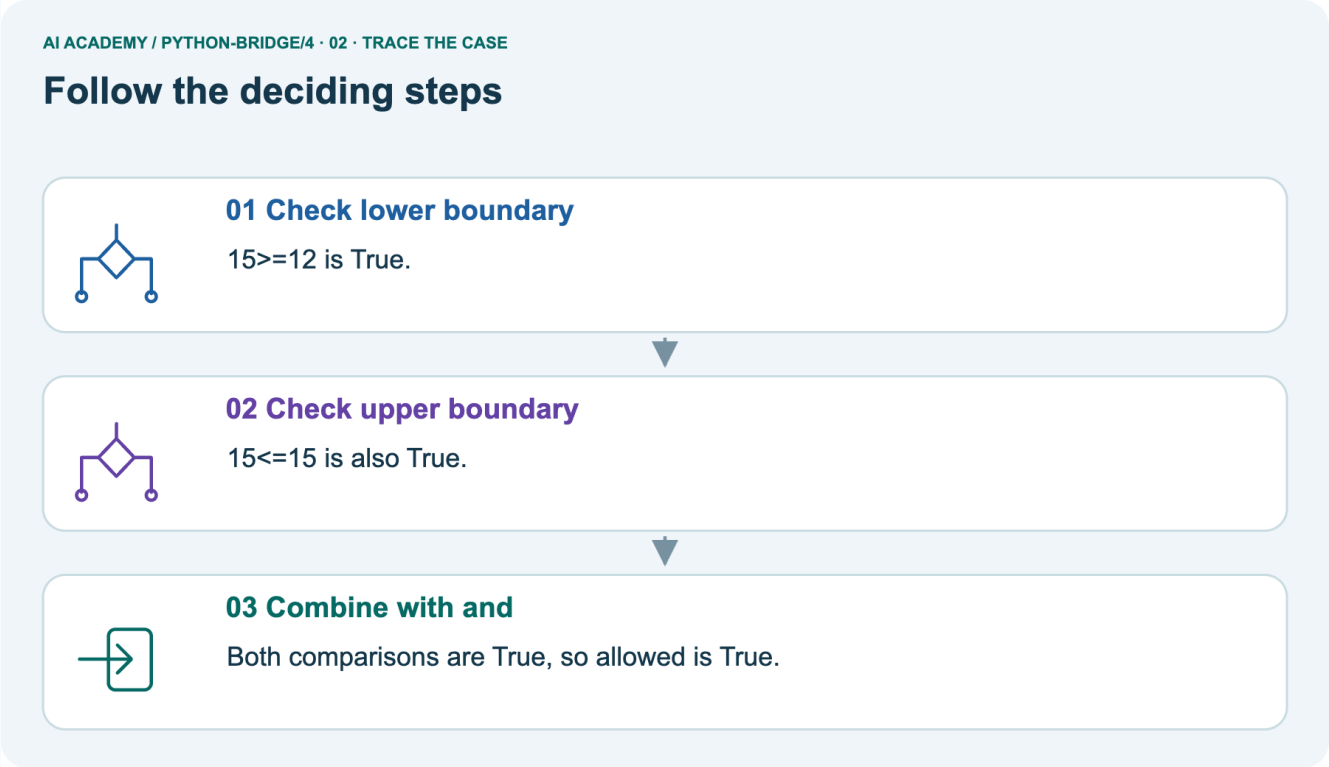
Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive. Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two requirements must both hold



Follow the labelled picture with the matching explanation. Open the vector diagram

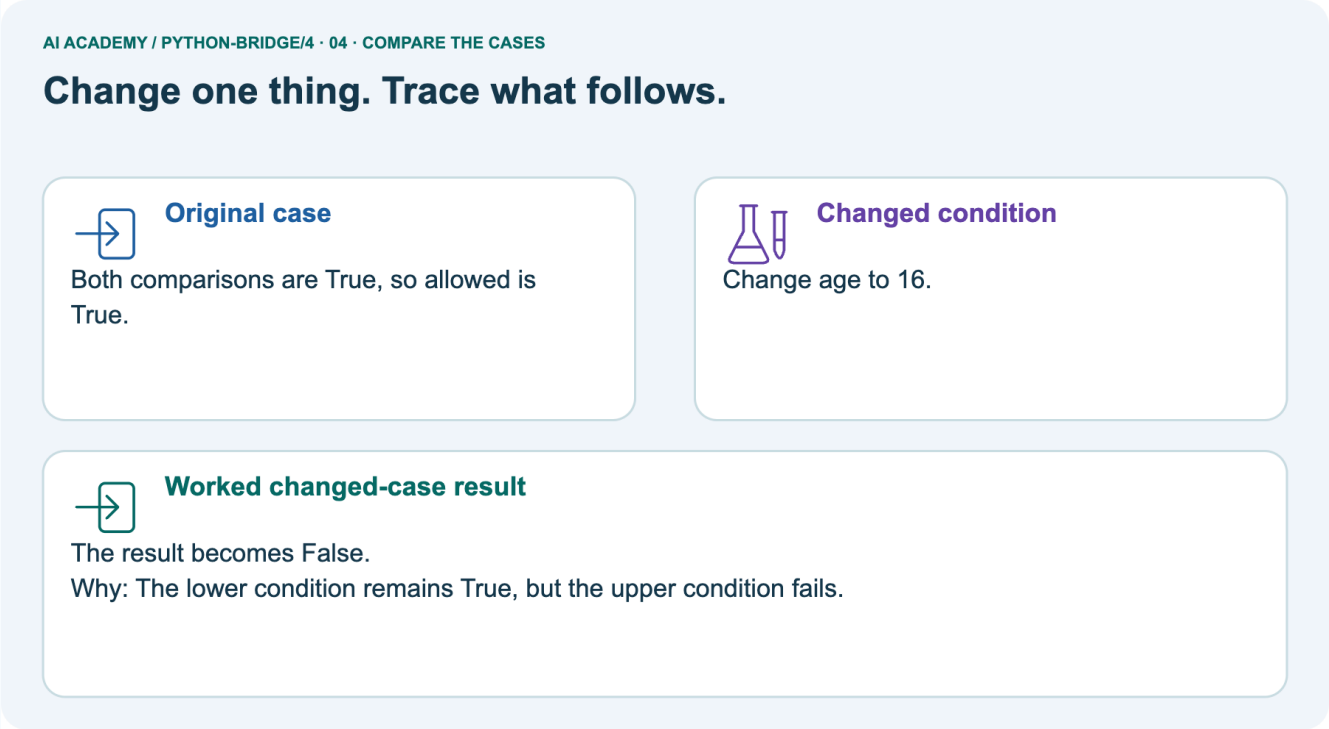
Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

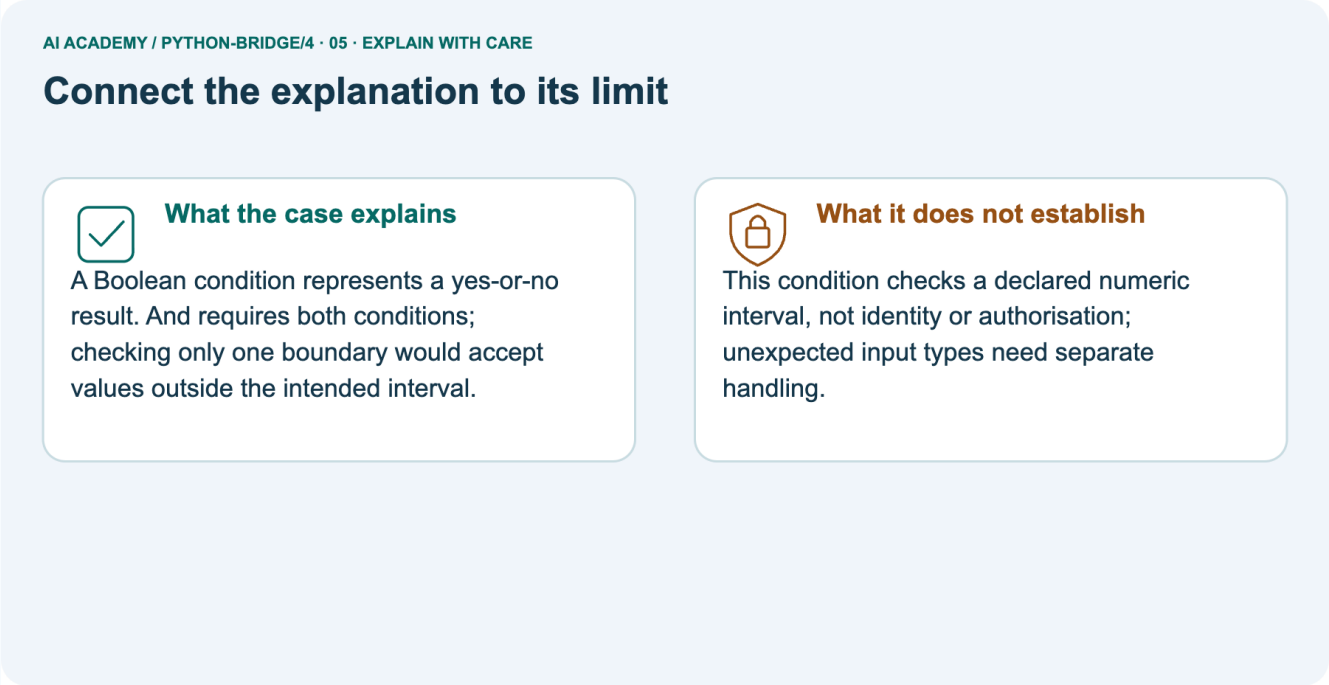
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The lower condition remains True, but the upper condition fails.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

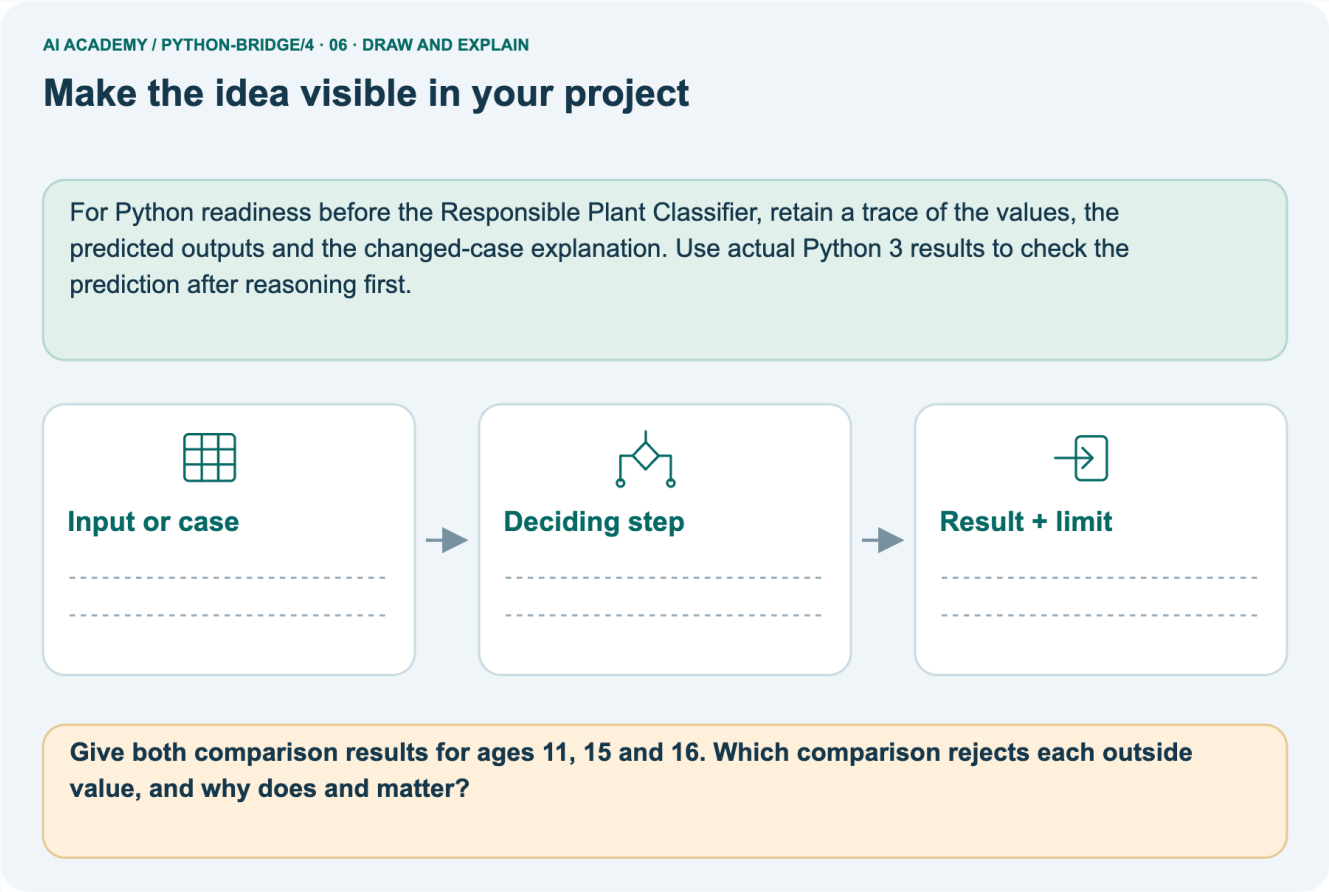
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Age 11 gives False/True, so False. Age 15 gives True/True, so True. Age 16 gives True/False, so False. Both interval boundaries must hold.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.